MILITARY SPECIFICATION

SHACKLE, BOMB, MARK 8 MOD 5

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for the manufacture, inspection and testing of Bomb Shackle Mark 8 Mod 5 which is used to carry and release weapons within the weight limits specified for the applicable aircraft.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-T-7743	Testing; Store Suspension Equipment, General Specification for
MIL-P-7936	Parts and Equipment, Preparation for Delivery
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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DRAWINGS

Bureau of Naval Weapons

63A119ASK1	List of Drawings, Bomb Shackle, Mark 8, Mod 5

2.2 Availability of Documents

(1) When requesting any of the applicable documents, refer to both title and number.

Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DC1, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The bomb shackles and hooks furnished under this specification shall be a product which has been inspected and passed the preproduction tests specified herein.

3.2 Parts and Materials - Parts and materials shall conform to applicable specifications and shall be as specified herein and on Drawing 63A119ASK1. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.2 Protective treatment - When materials are used in the construction of the bomb shackle that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.3 Design and construction - The design and construction of the bomb shackle shall be in accordance with the drawings listed on Drawing 63A119ASK1. The bomb shackle shall have two suspension hooks spaced 14 inches center to center to accept the type of lugs shown in Specification MIL-A-8591 for stores in the 1000 pound weight class.

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3.4 Performance - The bomb shackle shall pass the performance requirements specified in Section 4 when subjected to the applicable tests.

3.5 Identification of product - The bomb shackle shall be marked for identification in accordance with MIL-STD-130.

3.6 Workmanship - The bomb shackle shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The tests of the bomb shackle and hooks shall be classified as follows:

(a) Preproduction tests

(b) Acceptance Tests

4.3 Preproduction Tests - Preproduction tests shall consist of all the tests of this specification, except the operational test (4.6.2), and determination of the hook prestress load (see 6.4.6).

4.3.1 Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish preproduction samples for inspection to determine conformance with this specification. Quantity production shall be withheld until the preproduction samples have been pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these items from contract to contract, submission of further preproduction samples may be waived at the discretion of the procuring activity. The approval of preproduction samples or the waiving of the preproduction inspection shall not relieve the contractor of his obligation to submit samples for acceptance tests

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4.3.2 Preproduction samples shall consist of 3 bomb shackles and 25 additional hooks fabricated by quantity production methods and tools, except that the hooks shall not be prestressed (see 6.4.6). The samples shall be delivered for preproduction inspection to a Government testing laboratory as specified by the procuring activity (see 6.2).

4.4 Acceptance Tests -The acceptance tests shall consist of individual and sampling tests and shall be on a lot basis (see 4.4.2). All sampling tests will be performed by a Government testing laboratory as specified by the procuring activity (see 6.2).

4.4.1 Individual Tests - Each bomb shackle and hook on the contract or order shall be subjected to the following applicable tests Any bomb shackle or hook containing a defect shall be rejected.

(a) Examination (shackle) (4.6.1.1)

(b) Examination (hook) (4.6.1.2)

(c) Operational test (shackle) (4.6.2)

(d) Magnetic particle test (hook) (4.6.1.2.1)

4.4.2 Sampling Tests - A sample of bomb shackles and hooks shall be selected from each lot in accordance with the following:

TABLE I

SAMPLING CHART

LOT SIZE	SAMPLE SIZE	
	Complete Shackles	Hooks
Up to 99 Shackles	2	
100 - 300 Shackles	3	
Each 100 Hooks		3

The samples shall first have passed the individual tests and shall be subjected to the following inspections. If a sample fails to pass any of these tests, the lot represented by that sample shall be rejected. If a hook fails during a test of a bomb shackle, the lot of hooks (including shackles) represented by that sample shall be rejected.

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TABLE II

SAMPLING TESTS

TESTS	PARAGRAPH		SAMPLE
	SHACKLE	HOOK	NO.
Bomb shackle -			
Physical characteristics	4.6.3		1
Manual operation	4.6.3		1
Lag or time or release determination	4.6.3		1
Performance under icing conditions	4.6.3		1
Life test	4.6.3		1
Shock test	4.6.3		1
Vibration test	4.6.3		2
Static test	4.6.3		2(or 3)#
Hook -			
Ultimate load test		4.6.5	1
Dynamic load test		4.6.6	2 and 3

NOTE: Sample for tests marked # will be same as for vibration test, repaired as required. If another sample is available, it will be used for the static test.

4.4.2.1 Inspection lot -

4.4.2.1.1 Bomb shackles - Bomb shackles in quantities not greater than 300 and offered for delivery at one time shall be considered a lot for purposes of inspection.

4.4.2.1.2 Hooks - Hooks, whether delivered separately or as part of a bomb shackle, in quantities of 100 and heat treated together, shall be considered a lot for purposes of inspection.

4.5 Test conditions -

4.5.1 Hook test fixture - When specified in the applicable test, the hook shall be attached to suitable fixtures, one of which simulates the lug of Figure 1 of Specification MIL-A-8591 positioned in accordance with the following:

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(1) The lug-bearing surface on the toe of the hook is horizontal.

(2) The direction of the applied test load is normal to the lug-bearing surface.

(3) The fixture simulating the bomb lug shall be centered at a point located 0.437 inch horizontally from the hook pivot hole vertical center line.

(4) The restraining force is taken horizontally at the upper section of the hook that reacts the load produced by the lug.

4.6 Test methods -

4.6.1 Examination -

4.6.1.1 Bomb Shackle - The bomb shackle shall be thoroughly examined to insure that it has been properly assembled and adjusted and to determine conformance with this specification and applicable drawings with respect to all the req rements of material, workmanship, operational adequacy and reliability

4.6.1.2 Hook - The hook shall be thoroughly examined to determine conformance with this specification and applicable drawings with respect to all the requirements of material, workmanship, operational adequacy and reliability.

4.6.1.2.1 Magnetic particle inspection - In addition, each hook shall be subjected to the magnetic particle inspection in accordance with Specification MIL-I-6868. Special attention shall be given to the throat area and pivot hole area of the hooks. Both the longitudinal and circular magnetization methods shall be used. A current of 2,000 amperes minimum shall be applied for the circular magnetization method using a copper bar at the 0.375 inch hook pivot hole and the throat area. A 2,000 ampere current shall also be used for the longitudinal magnetization method. Magnetic particle inspection shall be performed prior to chromium plating, and also after chromium plating and prestressing.

4.6.1.3 The bomb shackle shall be examined to ascertain that the preparation for delivery conforms to this specification.

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4.6.2 Bomb shackle operational test - The completely assembled bomb shackle shall be cocked and latched (unloaded), a 2-1/4 inch, No. 5 taper pin inserted through the safety pin hole in the sideplates of the shackle, and the release lever of the shackle manually released. The release lever shall return to the latched position by its own spring action, thus recocking the shackle. The taper pin shall be removed, the shackle manually released and the hooks shall fully open. The travel and pull of the release lever for "release" and the travel of the release lever for "full stroke," shall be measured and shall conform to the "Shackle Release Data" table on Drawing 63A119F2. The above tests shall be repeated five times with the shackle carrying a 1000 pound load.

4.6.3 Store suspension equipment tests - The bomb shackle shall be subjected to and pass the following tests specified in Specification MIL-T-7743.

a. Physical characteristics
b. Manual operation
c. Lag or time of release determination
d. Performance under icing conditions
e. Life test
f. Shock test
g. Vibration test
h. Static test (as modified by 4.6.4)

4.6.4 Bomb shackle static test - The completely assembled bomb shackle shall be subjected to the following ultimate loads, each load to be applied separately in the direction noted:

20,000 pounds Vertically down
12,000 pounds Vertically up
9,500 pounds Longitudinally forward
9,500 pounds Longitudinally aft

4.6.5 Hook ultimate load test - The hook shall be attached to the test fixture specified in 4.5.1 and subjected to a vertically down load to destruction. The hook shall not fail below the ultimate load of 13,000 pounds.

4.6.6 Hook dynamic load test - The hook shall be attached to the test fixture as described in 4.5.1. The hook shall be subjected to 25,000 dynamic loading cycles with loads beginning at 2,250 pounds, increasing to 7,750 pounds and reduced to 2,250 pounds during each cycle. The cyclic rate shall be 1800 or 3600 cycles per minute. The load-time curve shall be equivalent to that produced by the Sonntag fatigue testing machine. The hook shall not fail below 25,000 cycles.

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4.7 Rejections -

4.7.1 Quality conformance inspection lot sample - The failure of any inspection lot sample to pass the inspections outlined in this specification shall cause tentative rejection of all shackles or hooks represented by that sample, subject to retest as hereinafter provided.

4.7.2 Individual rejection - The failure of any shackle, any hook, any piece, sub-assembly, or assembly of same to pass the individual inspections outlined in 4.4.1 shall cause its tentative rejection, subject to retest as hereinafter provided.

4.7.3 Salvaged pieces - Any piece(s) from a rejected shackle may be salvaged for use providing each salvaged piece is inspected, and is found to be in complete compliance with the requirements of this specification and the applicable drawings.

4.8 Reinspection -

4.8.1 Inspection lot samples - first retest - In the event any sample representing an inspection lot fails to pass the inspection outlined herein, the contractor shall have the opportunity of reworking the entire affected lot and resubmitting to retest twice as many samples, selected at random. Should all of these samples pass the required inspection, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.8.2 Inspection lot samples - second retest - If, in the contractor's opinion, the failure of the retest samples is due to a remediable defect, the contractor may, at the discretion of the Naval inspector, be offered the opportunity of reworking the entire affected lot and resubmitting to a second retest three times as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be finally rejected.

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4.8.3 Inspection subsequent to delivery - Any or all shackles may be subject to inspection for conformance with the requirements of this specification and applicable drawings after delivery to the Government.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level A - The bomb shackle shall be preserved and packaged in accordance with Specification MIL-P-7936, Level A. The packaging method shall conform to Specification MIL-P-116, Method III.

5.1.2 Level C - Preservation and packaging shall conform to Specification MIL-P-7936, Level C.

5.2 Packing -

5.2.1 Levels A, B, and C - The bomb shackle shall be packed in accordance with Specification MIL-P-7936, Level A, B, or C as specified in the procurement document.

5.3 Marking - Marking shall conform to Standard MIL-STD-129.

6. NOTES

6.1 Intended use - The bomb shackle is intended for use in suspending and releasing bombs, depth charges, etc., of a nominal weight between and including 100 pounds and 1500 pounds, on various types of aircraft. The bomb shackle may be actuated electrically by a bomb shackle release or by the manual release system of the airplane, and has provisions for easy individual self latching hooks, and safe locking when suspended or mounted, either externally, as from an airplane wing or fuselage, or internally, as within a bomb-bay.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging, and packing required
(c) Whether preproduction test is required (see 4.3.1.1)
(d) Where the preproduction test is to be conducted (see 4.3.1.2)
(e) Samples subjected to preproduction test are not to be considered or shipped as part of the contract or order

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(f) Samples subjected to sampling test are not to be considered or shipped as part of the contract or order (see 4.4.2)

6.3 Availability of dummy stores for testing - The contractor should submit requests to the procuring agency for dummy and inert stores to be used for testing as required by this specification.

6.4 Hook processing information - The following information is presented as a recommended guide to the manufacturer. This information is based on test and experimentation conducted under direction of the Government for the purpose of determining a satisfactory manufacturing process for hooks. The contractor may use this information at his discretion, but compliance with the procedures set forth below will not necessarily insure acceptance of the finished hooks. Acceptance of the hooks will be contingent upon satisfactory performance and test results as specified herein.

6.4.1 Forging - The forging-furnace conditions should be such as to give a minimum of decarburization. In the first heats obtained, the extent of decarburization should be determined in order that adequate allowances may be made for completely grinding off all decarburization resulting both from forge and heat treatment. Adequate stock should be retained on the forging blank to permit the removal of at least 0.010 inch of stock from all surfaces.

6.4.2 Heat treatment - The forgings should be heat-treated following the forging operation. The heat treatment should be in accordance with Specification MIL-H-6875 except as modified and amplified herein.

6.4.2.1 Furnace - The furnace or furnaces used for normalizing, annealing, and hardening preferably should be of the continuous type with an adjustable speed conveyor and should be operated through automatic temperature controllers to maintain the temperature of all points in the working zone within the required temperature ranges. The furnace used for tempering should be adequate to heat uniformly all parts of the furnace atmosphere. All furnaces should be the controlled-atmosphere type in order to prevent decarburization.

6.4.2.2 Loading - During normalizing, annealing, hardening, and tempering, the parts should be loaded in a manner so as to permit uniform air circulation around all surfaces of the hook and to permit uniform heating of all hooks in the load.

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6.4.2.3 Heating - The time required at temperature for the various operations should begin when the forgings and the furnace work zone have reached the required temperature. The temperature of all points in the work zone should be within ± 10°F of the temperature required for the parts, and should not fluctuate more than 10°F throughout the heating operation.

6.4.2.4 Normalizing - Following the forging operation, the parts should be normalized in accordance with Specification MIL-H-6875.

6.4.2.5 Annealing - Following the normalizing operation, the parts should be annealed in accordance with Specification MIL-H-6875.

6.4.2.6 Hardening - Following the annealing operation, the parts should be hardened in accordance with Specification MIL-H-6875.

6.4.2.7 Tempering - Immediately following the hardening process, the parts should be tempered. Experience with certain engineering steels indicate that for these steels the parts should be cooled after tempering and oil quenched. The hardness measurements should be made in the area which is not highly stressed when the hook is loaded.

6.4.3 Scale removal - Scale should be removed following the tempering process. Removal of scale should be performed by sand blasting or other mechanical method unless pickling or electrolytic methods are specifically approved by the procuring agency.

6.4.4 Machining - Care should be exercised during the machining of the hooks to prevent irregularities and tool marks on the hooks, especially in the throat area of the hooks under the 0.375 inch hole. Excessive tool marks on the hook may be cause for rejection.

6.4.5 Grinding - Grinding practice should be as used for tool steel grinding. A suitable coolant should be used during grinding. All grinding should be longitudinal to the hook, in no case across the hook. Grinding in the throat area should follow the curve of the hook on all surfaces. Checks and cracks due to grinding may be cause for rejection of the hook.

6.4.6 Prestressing - Experience has indicated that prestressing is essential to obtain the required fatigue life. In order to secure a substantial increase in fatigue life, it is recommended that all hooks be prestressed by cold stretching as illustrated in Figure 1. This prestressing operation benefits the hook by cold working the outer surface of the metal and also by placing the surface fibers under residual compressive stress. The prestress operation should be applied to the hook for a minimum of 60 seconds and will be the final operation in the manufacture of the hooks.

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6.4.6.1 Prestress load - The required prestress load will be determined during the Government conducted tests of the preproduction samples (see 4.3). In general, this load will be approximately 75 per cent of the ultimate strength of the hook. All hooks subsequent to the preproduction samples will be prestressed by the contractor.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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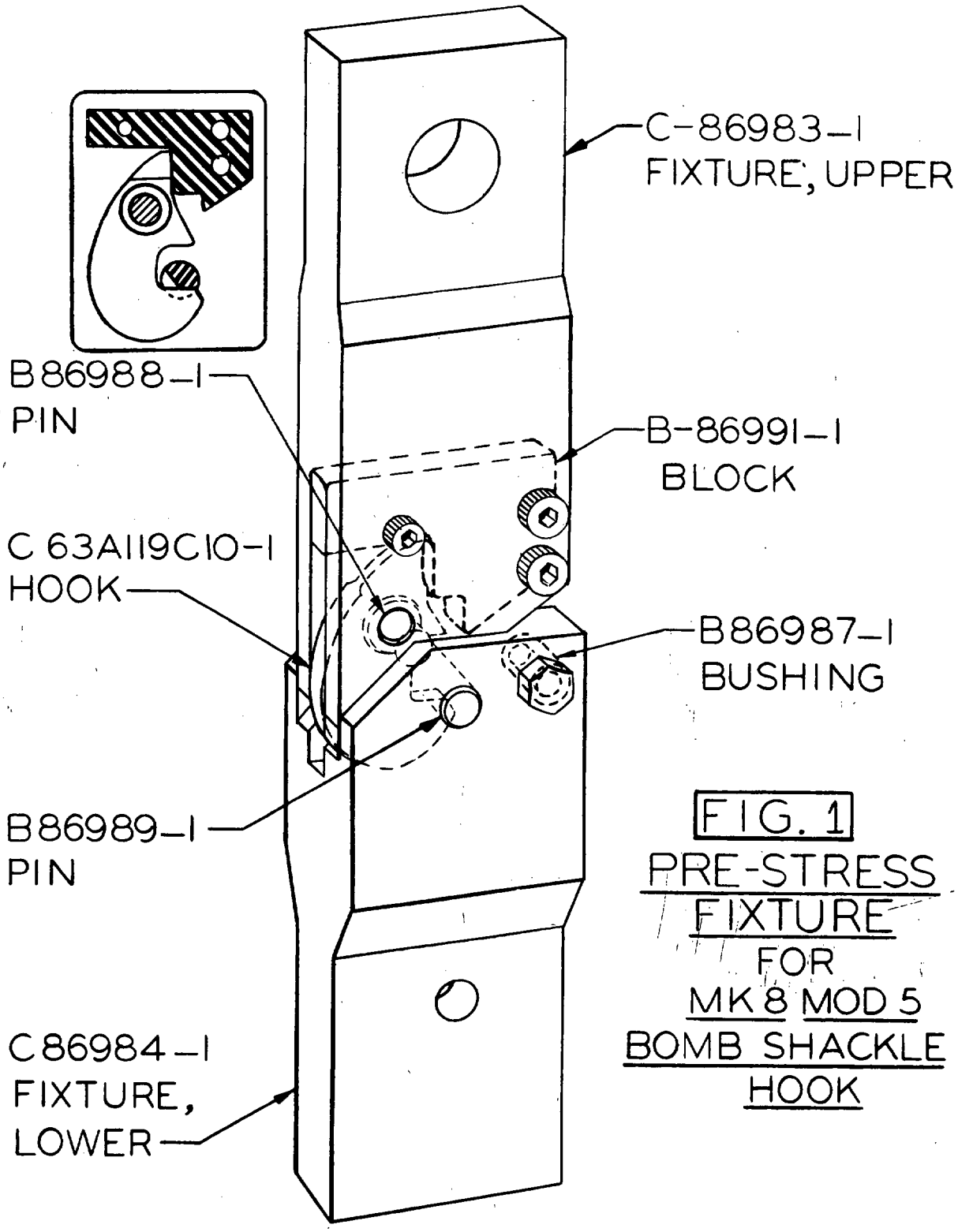


FIG. 1
PRE-STRESS FIXTURE FOR MK 8 MOD 5 BOMB SHACKLE HOOK

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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